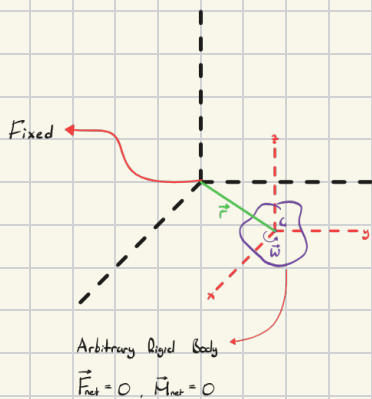


Intermediate Axis Theorem





$$\Rightarrow \vec{F} = m\vec{a}, \quad \vec{F} = m \frac{d\vec{v}}{dt}, \quad \vec{F} = \frac{d\vec{p}}{dt}$$

$$\Rightarrow (\vec{F} = m\vec{a}, \quad \vec{F} = m \frac{d\vec{v}}{dt}, \quad \vec{F} = \frac{d\vec{p}}{dt}) \times \vec{r}$$

$$\vec{M} - \frac{d\vec{H}}{dt} = I \frac{d\vec{\omega}}{dt}$$

For Inertial Frame of Reference

$$\vec{M} - \frac{d\vec{H}}{dt} = I \frac{d\vec{\omega}}{dt} = I \vec{\dot{\omega}}$$

For Non-Inertial Frame of Reference (with Transport Theorem)

$$\vec{M}_{\text{fixed}} - \left(\frac{d\vec{H}}{dt} \right)_{\text{fixed}} = \left(\frac{d\vec{H}}{dt} \right)_{\text{body}} + \vec{\omega} \times \vec{H}$$

In our case $\rightarrow \vec{M}_{\text{fixed}} = 0$ then,

$$\left(\frac{d\vec{H}}{dt} \right)_{\text{body}} + \vec{\omega} \times \vec{H} = 0, \quad I \vec{\dot{\omega}} + \vec{\omega} \times I \vec{\omega} = 0 \quad \text{and by using vector calculus,}$$

$$(I_x \dot{\omega}_x \hat{i} + I_y \dot{\omega}_y \hat{j} + I_z \dot{\omega}_z \hat{k}) + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \omega_x & \omega_y & \omega_z \\ I_x \omega_x & I_y \omega_y & I_z \omega_z \end{vmatrix} = 0$$

$$\underline{I_1 \dot{\omega}_x} + \underline{I_2 \dot{\omega}_y} + \underline{I_3 \dot{\omega}_z} + (I_1 - I_2) \omega_y \omega_z + (I_2 - I_1) \omega_x \omega_z + (I_3 - I_1) \omega_x \omega_y = 0$$

$$\Rightarrow \begin{cases} \dot{\omega}_x = \frac{(I_2 - I_1)}{I_1} \omega_y \omega_z \\ \dot{\omega}_y = \frac{(I_3 - I_1)}{I_2} \omega_x \omega_z \\ \dot{\omega}_z = \frac{(I_1 - I_2)}{I_3} \omega_x \omega_y \end{cases}$$

First Order, Homogeneous, Coupled, Non-Linear, Ordinary Differential Equations

$$\Rightarrow I_x = 8.816, \quad I_y = 10.065, \quad I_z = 16.666$$

1. Case

2. Case

3. Case

$$\omega_x = 5, \quad \omega_y = 0.1, \quad \omega_z = 0.1, \quad \omega_y \omega_z = 0 \quad \omega_x = 0.1, \quad \omega_y = 5, \quad \omega_z = 0.1, \quad \omega_x \omega_z = 0 \quad \omega_x = 0.1, \quad \omega_y = 0.1, \quad \omega_z = 5, \quad \omega_x \omega_y = 0$$

$$\dot{\omega}_x = 0$$

$$\dot{\omega}_x = \frac{(I_2 - I_1)}{I_1} \omega_y \omega_z$$

$$\dot{\omega}_x = \frac{(I_2 - I_1)}{I_1} \omega_y \omega_z$$

$$\dot{\omega}_y = \frac{(I_3 - I_1)}{I_2} \omega_x \omega_z$$

$$\dot{\omega}_y = 0$$

$$\dot{\omega}_y = \frac{(I_3 - I_1)}{I_2} \omega_x \omega_z$$

$$\dot{\omega}_z = \frac{(I_1 - I_2)}{I_3} \omega_x \omega_y$$

$$\dot{\omega}_z = \frac{(I_1 - I_2)}{I_3} \omega_x \omega_y$$

$$\dot{\omega}_z = 0$$

$\Rightarrow$ Obtain 2 Second-Order, Linear, Homogeneous, Ordinary Differential Equations with Constant Coefficients

$$\ddot{\omega} + b\dot{\omega} + c\omega = 0$$

1. Case

$$w_x(t) = S$$

$$w_y(t) = 0,1 [\cos(1,6351t) + 2,96521 \sin(1,63521t)]$$

$$w_z(t) = 0,1 [\cos(1,63521t) - 0,365346 \sin(1,63521t)]$$

3. Case

$$w_x(t) = 0,1 [\cos(4,01856t) - 0,978157 \sin(4,01856t)]$$

$$w_y(t) = 0,1 [\cos(4,01856t) + 1,02233 \sin(4,01856t)]$$

$$w_z(t) = S$$

2. Case

$$w_x(t) = -0,0999993 e^{-1,60886t} (-2,11112 + \frac{2}{e^{90772t}})$$

$$w_y(t) = S$$

$$w_z(t) = 0,0821428 e^{-1,60886t} (2,11112 + \frac{2}{e^{90772t}})$$

$\Rightarrow$ Notice that at 2. Case, $\lim_{t \rightarrow \infty} (w_x(t)) = \lim_{t \rightarrow \infty} (w_z(t)) = \infty$ due to term $= e^{-1,60886t}$.